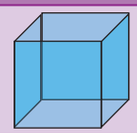
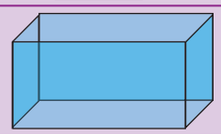
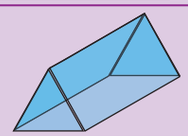
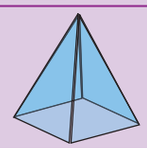
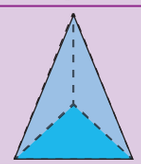
## Try these

Tabulate the number of faces (F), vertices (V) and edges (E) for the following polyhedrons. Also find F+V–E

| Solid | Name | F | V | E | F+V–E |
|---|---|---|---|---|---|
|  | Cube | 6 | 8 | 12 | |
|  | Cuboid | | | | |
|  | Triangular Prism | | | | |
|  | Square Pyramid | | | | |
|  | Triangular Pyramid | | | | |

What do you observe from the above table? We observe that, F+V–E = 2 in all the cases. This is true for any polyhedron and this relation F+V–E = 2 is known as **Euler's formula**.

### 2.4.2 Nets for building three dimensional (3-D) shapes

When we buy sweets, a shop keeper picks a flat shaped card which has some flips and makes a rectangular shaped box (cuboid) by folding it as shown in the figure. Then, he arranges the sweets in the box and gives it to us.

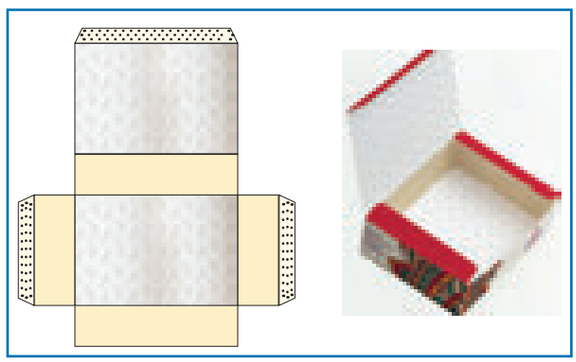
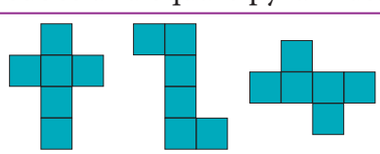
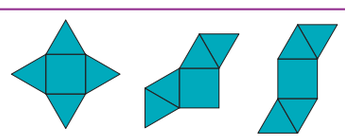


Fig. 2.35

The flat shaped card already designed for making the box excluding flaps (dotted lines) is known as a net.

For example, from the following nets we can build cubes and square pyramids.

| Nets for cubes | Nets for square pyramids |
|---|---|
|  |  |



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